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Ratio & Proportion

Test time on : Cryptarithmetic

QR CODE:

Ratio

- A **ratio** is a relationship between two numbers indicating how many times the first number contains the second.

- For **example**, if a bowl of fruit contains eight oranges and six lemons, then the **ratio** of oranges to lemons is eight to six (that is, 8∶6, which is equivalent to the **ratio** 4∶3).

Proportion

- A **proportion** is simply a statement that two **ratios** are equal. It can be written in two ways: as two equal fractions a/b = c/d; or using a colon, a:b = c:d. ...

- In problems involving **proportions**, we can use cross products to test whether two **ratios** are equal and form a **proportion**.

- A **proportion** is an equation with a **ratio** on each side. It is a statement that two **ratios** are equal. 3/4 = 6/8 is an **example** of a **proportion.**

Proportion

Now, let us assume that, in proportion, the two ratios are **a:b** & **c:d.**

The two terms **‘b’** and **‘c’** are called **‘means or mean term,’**

whereas the terms **‘a’** and **‘d’** are known as ‘**extremes or extreme terms.**’

$$a/b = c/d \text{ or } a : b :: c : d$$

Formula

1. If $u/v = x/y$, then $uy = vx$
2. If $u/v = x/y$, then $u/x = v/y$
3. If $u/v = x/y$, then $v/u = y/x$
4. If $u/v = x/y$, then $(u+v)/v = (x+y)/y$
5. If $u/v = x/y$, then $(u-v)/v = (x-y)/y$

Formula

If $u/v = x/y$, then $(u+v)/(u-v) = (x+y)/(x-y)$,

which is known as componendo -Dividendo Rule

If $u/v = v/x$, then $u/x = u^2/v^2$

If $u/v = x/y$, then $u = x$ and $v = y$

If $a/(b+c) = b/(c+a) = c/(a+b)$ and $a+b+c \neq 0$, then $a = b = c$

Question: 01

Are the ratios 4:5 and 8:10 said to be in Proportion?

A. YES
B. NO

Answer: A

Explanation: 01

4:5= 4/5 = 0.8 and 8: 10= 8/10= 0.8

Since both the ratios are equal, they are said to be in proportion.

Question: 02

Are the two ratios 8:10 and 7:10 in proportion?

A. YES
B. NO

Answer: B

Explanation: 02

8:10= 8/10= 0.8 and 7:10= 7/10= 0.7
Since both the ratios are not equal, they are not in proportion.

Question: 03

Given ratios are: a:b = 2:3, b:c = 5:2, c:d = 1:4 Find a: b: c:d.

A. 10:15:6:24

B. 10:6:15:24

C. 6:15:10:24

D. 24:6:10:15

Answer: A

Explanation: 03

Multiplying the first ratio by 5, second by 3 and third by 6, we have
a:b = 10: 15
b:c = 15 : 6
c:d = 6 : 24
In the ratio's above, all the mean terms are equal, thus
a:b:c:d = 10:15:6:24

Question: 04

Divide Rs. 90 in ratio 1:2 between Ram and Karan.

A. Rs.30 and Rs.60
B. Rs.60 and Rs.30
C. Rs.45 and Rs.45
D. Rs.80 and Rs.10

Answer: A

Explanation: 04

There are two parts, 1 and 2, the sum of which is 3 parts. Hence among the 3 parts, Karan gets 2 and Ram gets 1.

Therefore for 90 Rs (considered equivalent to 3 parts here)

Karan’s share = $2/3 \times 90$ = Rs.60

Ram’s share = $1/3 \times 90$ = Rs.30

Question: 05

Two numbers are respectively 20% and 50% more than a third number.The ratio of the two numbers is ________.

A. 5:4

B. 4:5

C. 6:7

D. 7:6

Answer: B

Explanation: 05

Let the third number be *x*.

Then, first number=120% of x=120x/100=6x/5

Second number =150% of x=150x/100=3x/2

Ratio of first two numbers $\left(\frac{6x}{5} : \frac{3x}{2}\right) = 12x : 15x = 4 : 5.$

Question: 06

A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share?

A. Rs.500

B. Rs.1500

C. Rs.2000

D. None of these

Answer: C

Explanation: 06

Let the shares of A, B, C and D be Rs.5x,Rs.2x,Rs.4x,Rs.3x

Then 4x-3x=1000

X=1000

So.B's share=2*1000=2000

Question: 07

Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats?

A. 2:3:4

B. 6:7:8

C. 6:8:9

D. None of These

Answer: A

Explanation: 07

Originally, let the number of seats for Mathematics, Physics and Biology be5x,7x and 8x respectively.

Number of increased seats are (140% of 5x) ,(150% of 7x) and (175% of 8x)

(140/100 x 5x),(150/100 x 7x),(175/100 x 8x)

7x.21x/2 and 14x

The required ratio is 7x:21x/2:14x

14x:21x:28x=2:3:4

Question: 08

In a mixture 60 litres, the ratio of milk and water is 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is __________.

A. 20litres

B. 30litres

C. 40litres

D. 60litres

Answer: D

Explanation: 08

Quantity of milk = $\left(60 \times \frac{2}{3}\right)$ litres = 40 litres.

Quantity of water in it = (60- 40) litres = 20 litres.

New ratio = 1 : 2

Let quantity of water to be added further be x litres.

Then, milk : water = $\left(\frac{40}{20 + x}\right)$.

Now, $\left(\frac{40}{20 + x}\right) = \frac{1}{2}$

$\Rightarrow 20 + x = 80$

$\Rightarrow x = 60.$

$\therefore$ Quantity of water to be added = 60 litres.

Question: 09

The ratio of the number of boys and girls in a college is 7 : 8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?

A. 8:9

B. 17:18

C. 21:22

D. Cannot be determined

Answer: C

Explanation: 09

Originally, let the number of boys and girls in the college be 7x and 8x .

Their increased number is (120% of 7x),(110% of 8x)

(120/100 x7x) and (110/100 x 8x)

(42x/5) and (44x/5)

The required ratio is (42x/5:44x/5)=(21:22)

Question: 10

Salaries of Ravi and Summit are in the ratio 2 : 3. If the salary of each is increased by Rs. 4000, the new ratio becomes 40 : 57. What is Summit's salary?

A. Rs. 17,000

B. Rs. 20,000

C. Rs. 25,500

D. Rs. 38,000

Answer: D

Explanation: 10

Let the original salaries of Ravi and Sumit be Rs. $2x$ and Rs. $3x$ respectively.

Then, $\dfrac{2x + 4000}{3x + 4000} = \dfrac{40}{57}$

$\Rightarrow 57(2x + 4000) = 40(3x + 4000)$

$\Rightarrow 6x = 68{,}000$

$\Rightarrow 3x = 34{,}000$

Sumit's present salary = $(3x + 4000)$ = Rs.(34000 + 4000) = Rs. 38,000.

Question: 11

The sum of three numbers is 98. If the ratio of the first to second is 2 :3 and that of the second to the third is 5 : 8, then the second number is __________.

A. 20

B. 30

C. 48

D. 58

Answer:B

Explanation: 11

Let the three parts be A, B, C. Then,

$A : B = 2 : 3$ and $B : C = 5 : 8 = (5 \times 3/5) : (8 \times 3/5) = (3 : 24/5)$

$A:B:C = 2:3:24/5 = 10:15:24$

$B = (98 \times 15/49) = 30$

Question: 12

The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

A. 3:3:10

B. 10:11:20

C. 23:33:60

D. Cannot be determined

Answer: C

Explanation: 12

Let A = $2k$, B = $3k$ and C = $5k$.

$$\text{A's new salary} = \frac{115}{100} \text{ of } 2k = \left(\frac{115}{100} \times 2k\right) = \frac{23k}{10}$$

$$\text{B's new salary} = \frac{110}{100} \text{ of } 3k = \left(\frac{110}{100} \times 3k\right) = \frac{33k}{10}$$

$$\text{C's new salary} = \frac{120}{100} \text{ of } 5k = \left(\frac{120}{100} \times 5k\right) = 6k$$

$$\therefore \text{ New ratio } \left(\frac{23k}{10} : \frac{33k}{10} : 6k\right) = 23 : 33 : 60$$

Question: 13

If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?

A. 2:5

B. 3:7

C. 5:3

D. 7:3

Answer: C

Explanation: 13

Let 40% of A = $\frac{2}{3}$ B

Then, $\frac{40A}{100} = \frac{2B}{3}$

$$\Rightarrow \frac{2A}{5} = \frac{2B}{3}$$

$$\Rightarrow \frac{A}{B} = \left(\frac{2}{3} \times \frac{5}{2}\right) = \frac{5}{3}$$

$\therefore$ A : B = 5 : 3.

Question: 14

The fourth proportional to 5, 8, 15 is ______.

A. 18

B. 24

C. 19

D. 20

Answer: B

Explanation: 14

Let the fourth proportional to 5, 8, 15 be x

Then, 5:8:15:x

5x=(8x15)

X=24

Ans:24

Question: 15

Two number are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is _______.

A. 27

B. 33

C. 49

D. 55

Answer: B

Explanation: 14

Let the numbers be $3x$ and $5x$.

Then, $\dfrac{3x - 9}{5x - 9} = \dfrac{12}{23}$

$\Rightarrow 23(3x - 9) = 12(5x - 9)$

$\Rightarrow 9x = 99$

$\Rightarrow x = 11.$

$\therefore$ The smaller number = $(3 \times 11) = 33$.

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+91 78150 95095 | codemithra@ethnus.com | www.codemithra.com